# STAFF REVIEW · H446 014 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

## Do Now — source answers

1. The item at index 2 — the third item.

2. A counter, condition and update.

3. The number of rows.

4. 9.

## I Do — source worked steps

1. Start with the old solution — CODE:
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]); — It works. But it repeats the same pattern three times.

2. Replace the three row statements — CODE:
for (int row = 0; row < board.GetLength(0); row++)
{
Console.WriteLine(board[row,0] + " | " +
board[row,1] + " | " +
board[row,2]);
} — Keep the row output the same but allow row to change inside the loop.

3. Replace the repeated columns — CODE:
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[0,col]);
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine(); — Now each column is selected by col . After the third column, finish the row.

4. Combine both loops — CODE:
for (int row = 0; row < board.GetLength(0); row++)
{
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[row,col]);
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine();
} — Predict how many times Console.Write(board[row,col]) executes: 9 .

5. Refine into DrawBoard() — CODE:
public static void DrawBoard()
{
for (int row = 0; row < board.GetLength(0); row++)
{
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[row,col]);
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine();
}
} — This method assumes board is stored at class level. Call DrawBoard() from Main() . You can reuse it after every move in Lesson 015.

## We Do — guided CHECK

1. What did we discover?
The code prints three values separated by pipes on every row. Only the row index changes (0 → 1 → 2).

2. Write or predict
Complete the counter condition to visit rows 0, 1 and 2 in sequence.
CODE:
for (int row = 0; row < ?; row++)
What did we discover?
CODE:
for (int row = 0; row < board.GetLength(0); row++)
Using the row count is better than assuming every board has three rows.

3. Write or predict
For one row, write a loop that visits every column. Which dimension returns the number of columns?
CODE:
for (int col = 0; col < ?; col++)
What did we discover?
CODE:
for (int col = 0; col < board.GetLength(1); col++)
The inner loop visits the columns across a single row.

4. Write or predict
Draw a simple trace: which values of row and col are visited first, and how many cells are printed?
What did we discover?
CODE:
[0,0] → [0,1] → [0,2]
[1,0] → [1,1] → [1,2]
[2,0] → [2,1] → [2,2]
There are nine cell visits. The outer loop moves down, the inner loop moves across.

## You Do — source mark schemes

1. 1. Re-run the sequential board
Begin with yesterday's three separate row statements before making any changes. Confirm the expected output 1–9.
CODE:
```
Console.WriteLine(board[0,0] + " | " + board[0,1] + " | " + board[0,2]);
Console.WriteLine(board[1,0] + " | " + board[1,1] + " | " + board[1,2]);
Console.WriteLine(board[2,0] + " | " + board[2,1] + " | " + board[2,2]);
```

2. 2. Loop over the rows
Only the row index changes. Replace the three row statements with a loop:
CODE:
```
for (int row = 0; row < board.GetLength(0); row++)
{
Console.WriteLine(board[row,0] + " | " +
board[row,1] + " | " +
board[row,2]);
}
```
Check that output remains identical.

3. 3. Loop over the columns
Inside a row, the three column values are another repeated operation.
CODE:
```
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[0,col]);
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine();
```
The row index is still 0 while col changes 0,1,2.

4. 4. Full DrawBoard method and runnable program
Place the board at class level and call DrawBoard() from Main() .
CODE:
```
using System;
public class Program
{
static string[,] board = {
{ "1", "2", "3" },
{ "4", "5", "6" },
{ "7", "8", "9" }
};

public static void DrawBoard()
{
for (int row = 0; row < board.GetLength(0); row++)
{
for (int col = 0; col < board.GetLength(1); col++)
{
Console.Write(board[row,col]);
```

```
if (col < board.GetLength(1)-1)
Console.Write(" | ");
}
Console.WriteLine();
}
}

public static void Main()
{
DrawBoard();
}
}
```

Expected: three rows; nine squares printed once. The method can be called after every move in Lesson 015.

### Exit Ticket — source answer

1. True. For a rectangular C# array, GetLength(1) gives its size in the second dimension: columns.